

Differentiable functions – 8

Smoothness and pathologies

Homework

1 сентября 2026 года

Theoretical minimum. Darboux' theorem; the discontinuity set of a derivative; Borel's theorem on prescribed Taylor coefficients; flat functions and the failure of analyticity; the Weierstrass nowhere differentiable function; Rademacher's and Lebesgue's theorems on almost everywhere differentiability (statements); the relation between C^∞ and real-analytic functions.

1. Prove that the set of points of discontinuity of a derivative is meagre, and give an example of a derivative discontinuous on a dense set.
2. Prove that a monotone function is differentiable almost everywhere (Lebesgue's theorem may be quoted) and construct a strictly increasing function with zero derivative almost everywhere.
3. Prove that $f(x) = e^{-1/x^2}$ for $x \neq 0$, $f(0) = 0$, is C^∞ with all derivatives vanishing at 0, and deduce that it is not real-analytic at the origin.
4. Prove that a C^∞ function with $|f^{(n)}(x)| \leq C^n n!$ on an interval is real-analytic there, and show that the bound is sharp.
5. Construct a C^∞ partition of unity subordinate to a finite cover of a segment, and explain why this is impossible in the analytic category.
6. Prove that there is a continuous function on $[0, 1]$ which is not differentiable at any point of a prescribed set of measure zero, and state Rademacher's theorem for Lipschitz functions.
7. Prove that if f is differentiable on $\mathbb{R}$ and f' is monotone, then f' is continuous, and deduce that a convex function has a continuous derivative wherever it is differentiable.

8. Give an example of a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ whose derivative is not Riemann integrable on $[0, 1]$, and explain why this does not contradict the Newton–Leibniz formula.
9. Prove that a function which is differentiable at every point of a segment and whose derivative is bounded is Lipschitz, and construct a differentiable function which is not Lipschitz on any interval.
10. Prove that the set of functions in $C[0, 1]$ which are differentiable at at least one point is meagre, and compare this statement with the corresponding one for monotone functions.